

Suggested problems: 1-11 odd

Problem 1: Verify that the probability function of a random variable $X \sim \mathcal{Bin}(n, p)$ really is a valid probability function.

Problem 2: Suppose that a discrete random variable X has probability function

$$f(x) = \begin{cases} \frac{c}{2^x} & : x = 0, 1, 2, \dots, \\ 0 & : \text{otherwise,} \end{cases}$$

for some positive constant c .

(a) Find the value of c .

(b) Find the probability of the event $C = \{0, 3, 83\}$.

(c) Find the probability of the event $D = \{0, 2, 4, 6, \dots\}$.

Problem 3: Verify that the probability function of a random variable $X \sim \mathcal{Uinf}(a, b)$ really is a valid probability function.